7.4

Integration of Rational Functions by Partial Fractions

In this section, we will learn:

How to integrate rational functions

by reducing them to a sum of simpler fractions.

- **Rational function**

it is an expression of a polynomial / a polynomial just like

- $\int \frac{x^3+x^2-3}{x^4+7x^2} dx$
- $\int \frac{1}{x^2-x-4} dx$

Example:

simplify (taking a common denominator)

$$\left(\frac{2}{x-1}-\frac{1}{x+2}\right)$$

Solution:

$$\left(\frac{2}{x-1}-\frac{1}{x+2}\right)=\frac{x+5}{x^2+x-2}$$

- **If we reverse the procedure:**

$$\frac{x+5}{x^2+x-2} = \left(\frac{2}{x-1} - \frac{1}{x+2}\right)$$

INTEGRATION BY PARTIAL FRACTIONS

- **integrate the function on the right side of this equation:**

$$\int \frac{x+5}{x^2+x-2}\,dx = \int \left(\frac{2}{x-1} - \frac{1}{x+2} \right) dx$$

$$= 2\ln|x-1| - \ln|x+2| + C$$

- Example
- Find

$$\int \frac{x-4}{x^2+2x-15}\,dx$$

- Example
- Find

$$\int \frac{1}{x^2-4}\, dx$$

- Example
- Find

$$\int \frac{x^2+9}{(x^2+4)\,(x^2-1)}\,dx$$

PARTIAL FRACTIONS

Example 1

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$$\int \frac{x^3 + x}{x - 1} dx$$

– **The degree of the numerator is greater than that of the denominator.**

Remark

You should a ware of the following points;

- If the degree of num.< The degree of denom.

  ⟶ (we will use partial fraction)
- If the degree of num.> The degree of denom.

  ⟶ (we will use Long division)



Table 8-1 The Four Cases for Setting Up Partial Fractions

Case	*Example*	*As Partial Fractions*
Case #1: Distinct linear factors	$\frac{x}{(x+4)(x-7)}$	$\frac{A}{x+4}+\frac{B}{x-7}$
Case #2: Distinct irreducible quadratic factors	$\frac{8}{(x^2+3)(x^2+9)}$	$\frac{A+Bx}{x^2+3}+\frac{C+Dx}{x^2+9}$
Case #3: Repeated linear factors	$\frac{2x+2}{(x+5)^2}$	$\frac{A}{x+5}+\frac{B}{(x+5)^2}$
Case #4: Repeated quadratic factors	$\frac{x^2-2}{(x^2+6)^2}$	$\frac{A+Bx}{x^2+6}+\frac{C+Dx}{(x^2+6)^2}$

PARTIAL FRACTIONS

Example 1

$$\int \frac{x^3 + x}{x - 1}\, dx = \int \left(x^2 + x + 2 + \frac{2}{x - 1} \right) dx$$

$$= \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2\ln|x - 1| + C$$

Example:

- **These constants can be determined as in the following:**

$$\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x}\, dx$$

Example 2

We factor the denominator as:

$$2x^3 + 3x^2 - 2x = x(2x^2 + 3x - 2)$$

$$= x(2x - 1)(x + 2)$$

It has three distinct linear factors.

So, the partial fraction decomposition of the integrand has the form

$$\frac{x^2 + 2x - 1}{x(2x-1)(x+2)} = \frac{A}{x} + \frac{B}{2x-1} + \frac{C}{x+2}$$

PARTIAL FRACTIONS

Example 4

$$\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}\,dx$$

- **The first step is to divide.**
- **The result of long division is:**

$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}$$

$$= x + 1 + \frac{4x}{x^3 - x^2 - x + 1}$$

Example 4

- The second step is to factor the denominator $Q(x) = x^3 - x^2 - x + 1$.

  – Since $Q(1) = 0$, we know that $x - 1$ is a factor, and we obtain:

$$\begin{aligned} x^3 - x^2 - x + 1 &= (x-1)(x^2-1) \\ &= (x-1)(x-1)(x+1) \\ &= (x-1)^2(x+1) \end{aligned}$$

Example 4

- The linear factor $x - 1$ occurs twice.
- So, the partial fraction decomposition is:

$$\frac{4x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

PARTIAL FRACTIONS

E. g. 4—Equation 8

- Multiplying by the least common denominator, $(x-1)^2(x+1)$, we get:

$$4x = A(x-1)(x+1) + B(x+1) + C(x-1)^2$$

$$= (A+C)x^2 + (B-2C)x + (-A+B+C)$$

Example 4

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$$A+C=0$$

$$B-2C=4$$

$$-A+B+C=0$$

PARTIAL FRACTIONS

Example 4

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- $A = 1$
- $B = 2$
- $C = -1$

PARTIAL FRACTIONS

Example 4

$$\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}\,dx$$

$$= \int \left[x + 1 + \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1} \right] dx$$

$$= \frac{x^2}{2} + x + \ln|x-1| - \frac{2}{x-1} - \ln|x+1| + K$$

$$= \frac{x^2}{2} + x - \frac{2}{x-1} + \ln\left|\frac{x-1}{x+1}\right| + K$$

CASE 3

- For instance, the function given by $f(x) = x/[(x - 2)(x^2 + 1)(x^2 + 4)$ has a partial fraction decomposition of the form

$$\frac{x}{(x-2)(x^2+1)(x^2+4)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{x^2+4}$$

CASE 3

Formula 10

- The term in Formula 9 can be integrated by completing the square and using the formula

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$$

PARTIAL FRACTIONS

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Example 5

$$\int \frac{2x^2 - x + 4}{x^3 + 4x}\,dx$$

– As $x^3 + 4x = x(x^2 + 4)$ can't be factored further, we write:

$$\frac{2x^2 - x + 4}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

PARTIAL FRACTIONS

Example 5

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$$2x^2 - x + 4 = A(x^2 + 4) + (Bx + C)x$$

$$= (A + B)x^2 + Cx + 4A$$

PARTIAL FRACTIONS

Example 5

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$A + B = 2$ $C = -1$ $4A = 4$

– Thus, $A = 1$, $B = 1$, and $C = -1$.

PARTIAL FRACTIONS

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Example 5

$$\int \frac{2x^2 - x + 4}{x^3 + 4x}\,dx = \int \left(\frac{1}{x} + \frac{x - 1}{x^2 + 4} \right) dx$$

PARTIAL FRACTIONS

Example 5

- In order to integrate the second term, we split it into two parts:

$$\int \frac{x-1}{x^2+4}\,dx = \int \frac{x}{x^2+4}\,dx - \int \frac{1}{x^2+4}\,dx$$

PARTIAL FRACTIONS

Example 5

- We make the substitution $u = x^2 + 4$ in the first of these integrals so that $du = 2x\,dx$.

PARTIAL FRACTIONS

Example 5

- We evaluate the second integral by means of Formula 10 with $a = 2$:

$$\int \frac{2x^2 - x + 4}{x(x^2+4)}\,dx$$

$$= \int \frac{1}{x}\,dx + \int \frac{x}{x^2+4}\,dx - \int \frac{1}{x^2+4}\,dx$$

$$= \ln|x| + \tfrac{1}{2}\ln(x^2+4) - \tfrac{1}{2}\tan^{-1}(x/2) + K$$

PARTIAL FRACTIONS

Example 6

$$\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3}\,dx$$

- The degree of the numerator is not less than the degree of the denominator.

- So, we first divide and obtain:

$$\frac{4x^2 - 3x + 2}{4x^2 - 4x + 3}$$

$$= 1 + \frac{x - 1}{4x^2 - 4x + 3}$$

Example 6

- Notice that the quadratic $4x^2 + x + 3$ is irreducible because its discriminant is $b^2 - 4ac = -32 < 0$.
  - This means it can't be factored.
- So, we don't need to use the partial fraction technique.

Example 6

- To integrate the function, we complete the square in the denominator:

$$4x^2 - 4x + 3 = (2x - 1)^2 + 2$$

  - This suggests we make the substitution $u = 2x - 1$.
    - Then, $du = 2\,dx$, and $x = \frac{1}{2}(u + 1)$.

PARTIAL FRACTIONS

Example 6

$$\int \frac{4x^2-3x+2}{4x^2-4x+3}\,dx$$

$$= \int \left(1+\frac{x-1}{4x^2-4x+3}\right)dx$$

$$= x+\tfrac{1}{2}\int \frac{\frac{1}{2}(u+1)-1}{u^2+2}\,du$$

$$= x+\tfrac{1}{4}\int \frac{u-1}{u^2+2}\,du$$

Example 6

$$= x + \tfrac{1}{4}\int \frac{u}{u^2+2}\,du - \tfrac{1}{4}\int \frac{1}{u^2+2}\,du$$

$$= x + \tfrac{1}{8}\ln(u^2+2) - \frac{1}{4}\cdot\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{u}{\sqrt{2}}\right) + C$$

$$= x + \tfrac{1}{8}\ln(4x^2-4x+3) - \frac{1}{4\sqrt{2}}\tan^{-1}\left(\frac{2x-1}{\sqrt{2}}\right) + C$$

NOTE

- Example 6 illustrates the general procedure for integrating a partial fraction of the form

$$\frac{Ax+B}{ax^2+bx+c} \quad \text{where} \quad b^2-4ac<0$$

NOTE

- We complete the square in the denominator and then make a substitution that brings the integral into the form

$$\int \frac{Cu+D}{u^2+a^2}\,du = C\int \frac{u}{u^2+a^2}\,du + D\int \frac{1}{u^2+a^2}\,du$$

  - Then, the first integral is a logarithm and the second is expressed in terms of $\tan^{-1}$.

CASE 4

a repeated irreducible quadratic factor.

CASE 4

- Suppose $Q(x)$ has the factor
$$(ax^2 + bx + c)^r$$
where $b^2 - 4ac < 0$.

CASE 4

Formula 11

- Then, instead of the single partial fraction (Formula 9), the sum

$$\frac{A_1x+B_1}{ax^2+bx+c}+\frac{A_2x+B_2}{(ax^2+bx+c)^2}+\cdots+\frac{A_rx+B_r}{(ax^2+bx+c)^r}$$

- occurs in the partial fraction decomposition of $R(x)/Q(x)$.

CASE 4

- Each of the terms in Formula 11 can be integrated by first completing the square.

PARTIAL FRACTIONS

Example 7

- Write out the form of the partial fraction decomposition of the function

$$\frac{x^3 + x^2 + 1}{x(x-1)(x^2+x+1)(x^2+1)^3}$$

PARTIAL FRACTIONS

Example 7

We have:

$$\frac{x^3+x^2+1}{x(x-1)(x^2+x+1)(x^2+1)^3}$$

$$=\frac{A}{x}+\frac{B}{x-1}+\frac{Cx+D}{x^2+x+1}+\frac{Ex+F}{x^2+1}$$

$$+\frac{Gx+h}{(x^2+1)^2}+\frac{Ix+J}{(x^2+1)^3}$$

Example 8

Evaluate

$$\int \frac{1-x+2x^2-x^3}{x(x^2+1)^2}\,dx$$

The form of the partial fraction decomposition is:

$$\frac{1-x+2x^2-x^3}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$$

Example 8

Multiplying by $x(x^2 + 1)^2$, we have:

$$-x^3 + 2x^2 - x + 1$$

$$= A(x^2 + 1)^2 + (Bx + C)x(x^2 + 1) + (Dx + E)x$$

$$= A(x^4 + 2x^2 + 1) + B(x^4 + x^2) + C(x^3 + x) + Dx^2 + Ex$$

$$= (A + B)x^4 + Cx^3 + (2A + B + D)x^2 + (C + E)x + A$$

PARTIAL FRACTIONS

Example 8

- equa… …em
  …get… …system

$$A + B = 0$$

$$C = -1$$

$$2A + B + D = 2$$

$$C + E = -1$$

$$A = 1$$

– This has the solution
$A = 1$, $B = -1$, $C = -1$, $D = 1$, $E = 0$.

PARTIAL FRACTIONS

Example 8

$$\int \frac{1-x+2x^2-x^3}{x(x^2+1)^2}\,dx$$

$$= \int \left(\frac{1}{x} - \frac{x+1}{x^2+1} + \frac{x}{(x^2+1)^2} \right) dx$$

$$= \int \frac{dx}{x} - \int \frac{x}{x^2+1}\,dx - \int \frac{dx}{x^2+1} + \int \frac{x\,dx}{(x^2+1)^2}$$

$$= \ln|x| - \tfrac{1}{2}\ln(x^2+1) - \tan^{-1} x - \frac{1}{2(x^2+1)^2} + K$$

AVOIDING PARTIAL FRACTIONS

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partial fractions can be avoided when integrating a rational function.

AVOIDING PARTIAL FRACTIONS

$$\int \frac{x^2+1}{x(x^2+3)}\,dx$$

- For instance, the integral

- could be evaluated by the method of Case 3.

AVOIDING PARTIAL FRACTIONS

- However, it is much easier to observe that,

if $u = x(x^2 + 3) = x^3 + 3x$, then $du = (3x^2 + 3)\,dx$ and so

$$\int \frac{x^2+1}{x(x^2+3)}\,dx = \tfrac{1}{3}\ln|x^3+3x|+C$$

RATIONALIZING SUBSTITUTIONS

- Some nonrational functions can be changed into rational functions by means of appropriate substitutions.
  - In particular, when an integrand contains an expression of the form $\sqrt[n]{g(x)}$, then the substitution $u = \sqrt[n]{g(x)}$ may be effective.

RATIONALIZING SUBSTITUTIONS

Example 9

$$\int \frac{\sqrt{x+4}}{x}\,dx$$

- Let $u = \sqrt{x+4}$
- Then, $u^2 = x + 4$
- So, $x = u^2 - 4$ and $dx = 2u\,du$

RATIONALIZING SUBSTITUTIONS

Example 9

her

$$\int \frac{\sqrt{x+4}}{x}\,dx = \int \frac{u}{u^2-4}\,2u\,du$$

$$= 2\int \frac{u^2}{u^2-4}\,du$$

$$= 2\int \left(1+\frac{4}{u^2}-4\right)du$$

Example 9

We evaluate this integral by factoring $u^2 - 4$ as $(u - 2)(u + 2)$ and using partial fractions.

RATIONALIZING SUBSTITUTIONS

Example 9

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wi 2.

$$\int \frac{\sqrt{x+4}}{x}\,dx$$

$$= 2\int du + 8\int \frac{du}{u^2-4}$$

$$= 2u + 8\cdot\frac{1}{2\cdot 2}\ln\left|\frac{u-2}{u+2}\right| + C$$

$$= 2\sqrt{x+4} + 2\ln\left|\frac{\sqrt{x+4}-2}{\sqrt{x+4}+2}\right| + C$$